Alexander Amadeo-Ranch

Aerospace Engineering · Mechanical Design, Structures and Hardware Integration

Los Angeles Metropolitan Area | alexengineer.net | linkedin.com/in/alexamadeoranch | U.S. citizen

EDUCATION

University of California, Irvine Sep 2023 – Present

B.S. Aerospace Engineering, in progress GPA 3.759

Minor in Management, Paul Merage School of Business. Coursework ends March 23, 2027; available for full-time work from April 2027.

Relevant coursework: Mechanics of Structures and Lab, Lightweight Structures, Dynamics, Mechanical Systems Lab, Aerospace Lab, Introduction to Control Systems, Statics, Fluid Dynamics I, Thermodynamics, Materials Science, Electric Circuits, Engineering Computations (MATLAB), CubeSat I–III.

Honors: Dean's Honor List, multiple quarters; Sigma Gamma Tau, national aerospace engineering honor society (initiation in progress).

Monte Vista High School 2019 – 2023

High school diploma with honors; California Scholarship Federation, National Honor Society, eight-semester honor roll.

ENGINEERING EXPERIENCE

Boundary Remote Subsurface Solutions Los Angeles Metropolitan Area

UAV Hardware Engineering Intern Sep 2026 – Present

- Completed a system-level engineering assessment of the current UAV design covering mass and power budgets, structural load paths, thrust, propeller performance and the flight envelope, using SOLIDWORKS, ANSYS and COMSOL.
- Delivered the assessment two weeks ahead of its deadline; the findings informed a change in the company's hardware direction.
- Collaborated across departments and presented technical findings to nontechnical managers to support hardware decisions.

UCI CubeSat, Student spacecraft program Irvine, CA

Systems Engineer Mar 2026 – Present

- Built FlatSat V1, a modular aluminum-extrusion bench fixture that holds up to 12 avionics PCBs for ground integration and testing; simplifying the design reduced the total FlatSat purchase-order cost by 30% compared with V0.
- Designed, 3D-printed in PLA and installed all 14 custom extrusion mounts, replacing an expensive commercial part with a long lead time so assembly did not wait on procurement.

Structural Engineer Sep 2025 – Mar 2026

- Designed and evaluated structural components for internal bus integration and external radiation shielding under mass, stiffness and envelope constraints, using SOLIDWORKS and ANSYS Mechanical static-structural and modal analysis.
- Ran trade studies on manufacturability, integration tolerances and launch survivability, and produced documentation for fabrication, assembly and integration.
- Coordinated payload-interface work with the STMS research group.

Across roles: structural analysis, 2025–present

- Analyzed the M3 standoffs carrying the internal bus stack for end-cap-to-end-cap tension and launch vibration, considering PCB vibration limits; calculated a factor of safety of 5 against yield by calculation and simulation (physical vibration testing pending). Design approved by subteam leads and project managers.
- Designed to thermal constraints supplied by the thermal-analysis subteam and reviewed their documentation.

Chief Integration Officer Oct 2024 – Sep 2025

- Coordinated mechanical, electrical and payload interfaces across subteams during integration and served as the primary liaison between CubeSat leadership and the STMS research group.
- Contributed to internal and external communications on spacecraft integration and program status.

Executive Assistant Sep 2023 – Oct 2024

- Supported project leadership with documentation, scheduling and internal coordination during early program development.

ENGINEERING PROJECTS

Autonomous Robot Design and Verification, MAE 106, team of four, experimental lead Jan – Mar 2026

- Led duty-cycle and hinge-height experiments (five settings × 10 runs each) for a four-person pneumatic-robot team. Data informed the selected settings; the robot met its autonomous 3 ft / 90° turn / 3 ft requirement.
- Calibrated the magnetometer in MATLAB, measured heading noise and one-hour drift, and quantified servo position error (about 10° mean overshoot) before integration.

Mission to Didymos Transfer Analysis, MAE 146 Astronautics, course project Feb – Mar 2026

- Wrote MATLAB two-body propagation, element-conversion and Lambert functions; searched 16,290 departure/arrival pairs on a 201 × 201 grid, reducing modeled transfer Δv from 20.028 to 7.873 km/s within the search window.

Inverted-Pendulum Controller Design, MAE 106, individual assignment Feb 2026

- Derived closed-loop poles for a cart–pendulum under PD control and selected Kp = 50 and Kd = 5 (poles at −2.5 ± 5.39j); stable from initial angles up to 0.78 rad in simulation.

Heavy-Lift Recovery UAV Proposal (ExtrakPak), ENGR 190W, team of three Jan – Mar 2025

- Wrote the airframe, propulsion, grappling-system, remote-command and regulatory sections of a concept proposal with 200 kg payload and one-hour endurance targets, including thrust-to-weight feasibility estimates. Concept only; not built.

Reinforced Ice for Lunar Micrometeoroid Protection, ENGR 54, team of seven May – Jun 2025

- Helped plan and run drop tests on 33 reinforced ice samples and the energy-balance analysis for a modeled 3 cm, 42 km/s iron micrometeoroid; the calculated ice thickness was impractical.

Periodic Table Database, MAE 10, team project Sep – Dec 2023

- Built a MATLAB tool for element lookup and periodic-trend plots from a structured CSV dataset.

SOFTWARE, WEB AND AUTOMATION EXPERIENCE

Founder to Founder Partners San Francisco Bay Area

System Engineer Jul 2026 – Present

- Built an AI-assisted Python/Node.js research pipeline that turned licensing, Census, federal-contract, industry-directory and company data into 1,000 unique qualified lead records spanning all 50 states.
- Added verification and traceability controls: screening against a 1,193-record exclusion registry, maintaining 3,822 source-evidence records and documenting 154 disqualifications before delivery.
- Delivered the complete system in approximately 18 engineering hours (about 56 leads per hour), saving 30% of the client's budget with $0 in external operating costs.

Spirit Hero, LLC California

Systems & Automation Engineer Jun 2026 – Present

Software Developer Mar – Jun 2026

- Modernized Shopify theme UI/UX and collection pages while keeping the existing brand.
- Reduced page load time by 5–10% and front-end warnings by 20% through JavaScript cleanup, Liquid debugging and conservative theme refactors.
- Built a web parser that produced a 100,000+ record lead workbook, with an updatable extraction and search setup for repeat runs.

Sherman Oaks Chamber of Commerce Los Angeles, CA

Website Maintainer Oct 2026 – Present

Web Developer May – Oct 2026

- Built the Sherman Oaks 2027 centennial website end to end in HTML, CSS and vanilla JavaScript: one hub for four events, sponsorship outreach and resident participation.
- Built an interactive Leaflet/MapLibre map of 8 historic locations with 7 then-and-now photo comparisons, synchronized pins and shareable location links.
- Built sponsorship comparison and inquiry flows covering 20 offerings across the 4 events.
- Deferred more than 1 MB of mapping libraries and checked layouts from 320 to 1440 px; added technical SEO across 7 canonical pages and keyboard, focus and contrast fixes, reaching zero automated axe-core violations across 36 page-and-viewport combinations.

Amadeo & Associates California

Software Engineer Oct 2025 – Present

Automotive Technician Dec 2020 – Oct 2026

Administrative Assistant Aug 2014 – Dec 2020

SOFTWARE PROJECTS

AeroLab 2D and 3D, Personal, alexengineer.net/aerolab Published Aug 2026

- Built browser aerodynamics labs in JavaScript and WebGL 2: lattice-Boltzmann flow view, Hess–Smith panel method and integral boundary layer for airfoils; vortex-lattice aircraft analysis, Barrowman rocket stability and a re-entry trajectory model.

EXOHUNT, Personal, open source Jul 2026 – Present

- Building an open-source Python pipeline for public TESS light curves: BLS transit searches, a known-planet benchmark and catalog cross-checks; other signals are treated as unvetted candidates.

Read-Only OBD-II Diagnostic Dashboard, Personal Published May 2026

- Built a local FastAPI/WebSocket dashboard for live vehicle diagnostic data, restricted by design to an allowlist of read-only OBD-II and UDS services.

Slamcaster, Personal Aug 2025

- Built and maintained a multi-page static publishing site on GitHub Pages with a custom domain (180+ commits).

OPERATIONS, LEADERSHIP AND TEACHING

Target Irvine, CA

Closing Expert Aug 2026 – Present

Town of Danville, Aquatics Danville, CA

Pool Manager Aug 2022 – Sep 2024

- Supervised a staff of 75+ and up to 120+ patrons at a time; planned logistics for and supervised select events with 100+ attendees.

- Led responses to real (non-drill) facility emergencies.
- Wrote training documentation and manager report templates.

Senior Guard Aug 2021 – Aug 2022

- Supported facility operations, patron safety and emergency-response procedures.

Lifeguard Aug 2019 – Aug 2021

Junior Guard Jun 2016 – Aug 2019

Freelance California

Private Tutor Oct 2021 – Jun 2026

- Taught mathematics, physics and computer science one-on-one to students from middle school through college, with individualized lesson plans; some client relationships lasted up to three years.

FurFlex California

Founder May 2025 – May 2026

- Ran a small e-commerce business: finances, domestic supplier coordination for Walmart and Amazon sales, marketing, fulfillment and back-end web infrastructure; liquidated the infrastructure to Spirit Locker Organization.

Self-employed California

Private Swim Instructor May 2021 – Oct 2024

- Taught private swim lessons across ages and skill levels; managed scheduling, session planning and safety.

YouTube

YouTube Partner Apr 2019 – Oct 2023

- Produced educational space and science videos (scripting, editing, publishing), moderated the channel community and coordinated sponsorships.

SKILLS

Design and analysis: SOLIDWORKS (CSWA), ANSYS Mechanical (static structural, modal), COMSOL (UAV assessment), MATLAB, mass, power and load-path budgeting.

Fabrication and test: PLA 3D printing, mechanical assembly, soldering, laser cutting, bandsaw, automotive maintenance, Arduino sensor calibration and test logging.

Software: Python, Node.js, JavaScript, HTML/CSS, Shopify Liquid, WebGL, Git/GitHub.

Languages: English (native), Spanish (elementary).

CERTIFICATIONS AND TRAINING

Engineering and safety: Certified SOLIDWORKS Associate (CSWA), Oct 2025; Laboratory Safety (UC Learning Center), Apr 2026; Shop Safety (UC Learning Center), Oct 2025; Hazardous Waste (UC Learning Center), Oct 2025; Hazardous Materials Incidents Emergency Procedures (UC Learning Center), Oct 2025.

AI and cloud courses (Apr – Jun 2026): NVIDIA AI Infrastructure and Operations Fundamentals; AWS Generative AI and AI Agents with Amazon Bedrock; Microsoft AI & ML Engineering; Microsoft Generative AI Engineering; Meta Gen AI in Social Media Marketing; Google AI Professional Certificate.

Historical: American Red Cross Lifeguard (Town of Danville), issued Jun 2023, expired Jun 2025.

HONORS AND ACTIVITIES

Eagle Scout with Gold Palm, Scouts BSA, Apr 2022; Taekwondo first-degree black belt (1st Dan), 2016 (taught classes 2015–2016).